Frederik Morbe

Lead Mechanical Design Engineer | R&D

■ Mörfelden-Walldorf, Germany ■ +49 1731989100

PERSONAL

Location: Mörfelden-Walldorf, DE
Phone: +49 1731989100

LANGUAGES

German Native
English Full Professional
Mandarin Elementary (A2)

CORE COMPETENCIES

- Mechanical Design & CAD
- Medical Device Development
- Technical Project Mgmt.
- Design Transfer & Validation
- 3D Printing & Prototyping
- IQ / OQ / PQ Qualification
- Technical Documentation
- AI & Modern Technologies

IT & TECHNICAL TOOLS

Solid Edge ●●●●●
SolidWorks ●●●●○
KeyShot ●●●●●
MS Office ●●●●●
SAP / PLM ●●●●○
Cinema 4D ●●●○○
Python ●●●○○
Adobe Suite ●●●●○

METHODOLOGIES

- End-to-End Product Dev.
- Design Freeze & DHF
- Design Transfer
- IQ / OQ / PQ
- 3D Printing / FDM
- SAP PM / Pro.File PLM
- Agile / Scrum
- LLM / AI Integration

PROFESSIONAL SUMMARY

Mechanical design engineer with 10+ years developing clinical-grade medical research devices at Miltenyi Biotec. Proven expertise in CAD (Solid Edge, SolidWorks) and end-to-end product development, with key contributions to CliniMACS Prodigy and MACSima. Complements a strong engineering foundation with an international academic profile and active interest in Python and AI/LLM tools.

WORK EXPERIENCE

Mechanical Design Engineer – R&D

Miltenyi Biotec B.V. & Co. KG · Jan 2016 – Present

- Leads technical project management and mechanical design of novel medical research devices, overseeing all phases from concept through design freeze and production handover.
- Drove end-to-end development and testing of new research devices (2018–2021), managing design, qualification, and documentation in a regulated environment.
- Part of the R&D team for different Devices covering major system components, housings, and common modules shared across multiple devices.
- Utilises Solid Edge for 3D design and KeyShot for high-quality product rendering.
- Collaborated with cross-functional teams to ensure designs met functional, aesthetic, and production requirements.

Development Engineer (On-site: Miltenyi Biotec)

Brunel GmbH · Mar 2015 – Dec 2015

- Seconded as development engineer to Miltenyi Biotec, supporting mechanical design activities and gaining in-depth experience in medtech product development workflows.

Student Trainee & Bachelor Thesis – R&D

Thermo Fisher Scientific · May 2013 – Jan 2014

- Completed Bachelor's thesis within the R&D department, followed by continued engagement as an R&D working student.

KEY PROJECTS (MILTENYI BIOTEC)

Development of New Medical Devices · *Jan 2018 – Jan 2021*

- Full end-to-end development and qualification of two new medical research devices, from initial concept through design freeze and production handover.

CliniMACS Prodigy HMI · *Jan 2017 – Jan 2018*

- Mechanical design and development of the Display Unit for Miltenyi's CliniMACS Prodigy — a leading clinical cell therapy processing system.

MACSima Imaging System · *Jan 2017 – Jan 2018*

- Product design of major mechanical components including device housing, automatic door mechanisms, and integrated display units.

EDUCATION

B.Sc. Mechanical Engineering (Maschinenbau)

Frankfurt University of Applied Sciences · 2009 – 2013

- Core studies covering mechanical engineering, materials science, design methodology, and manufacturing processes.

Chinese (Mandarin) Language Studies

Wenzao Ursuline University of Languages, Taiwan · 2014

- Intensive language programme at a leading Asian languages university; achieved Basic Chinese Level A2 Certificate with a grade of 97/100.